

MediKiosk - Clinical Domain Research

SOCRATES History Structuring, Dashavidha Pariksha, and Clinical Note Formats

SIH 2026 Problem Statement #47 - Research compiled 2026-09-01

1. SOCRATES - Clinical History Structuring Framework

SOCRATES is an 8-field mnemonic used by physicians, nurses, and emergency services to systematically explore a patient's presenting symptom - most commonly pain, but adaptable to any symptom. It is the standard for eliciting a comprehensive History of Present Illness (HPI) and sits inside a broader interview sequence: open question -> SOCRATES gap-fill -> red-flag screens -> past medical history -> drugs -> allergies -> family/social history -> ICE (Ideas, Concerns, Expectations) -> summary back to patient.

Field	Probe	Clinical Purpose
S - Site	"Where is it? Can you point with one finger?"	Localises anatomy; diffuse pain suggests visceral
O - Onset	"When did it start? Sudden or gradual?"	Thunderclap (sudden severe) suggests SAH; gra
C - Character	"Sharp, dull, crushing, burning, tight, colicky?"	Often the highest-yield discriminator between cau
R - Radiation	"Does it spread anywhere?"	Cardiac pain to jaw/L arm; biliary to right scapula.
A - Associations	"What else happens with it?"	Nausea, sweating, fever, breathlessness narrow
T - Timing	"Constant or intermittent? How long per episode?"	Nocturnal, episodic, and progressive patterns eac
E - Exacerbating/Relieving	"What makes it better or worse?"	Exertional -> ischaemia; sitting forward -> pericar
S - Severity	"Out of 10 - and what does 10 look like for you?"	Functional impact matters as much as the numbe

SOCRATES adapts to non-pain symptoms: cough (throat vs chest, dry vs productive, day vs night), breathlessness (chest tightness vs general, gradual vs sudden, orthopnoea, paroxysmal nocturnal dyspnoea, leg swelling), headache (frontal/occipital/unilateral, thunderclap vs band-like, nausea, photophobia, neck stiffness), and palpitations (chest flutter, rest vs exertion, regular vs irregular).

2. Dashavidha Pariksha - The Tenfold Ayurvedic Examination

Dashavidha Pariksha (literally 'tenfold examination') is the classical Ayurvedic method of assessing the diseased person (atura/rogi) before treatment, described in Charaka Samhita, Vimana Sthana, Adhyaya 8. It is person-focused (unlike SOCRATES which is symptom-focused). The ten parameters jointly determine Rogi Bala (patient strength), Roga Bala (disease strength), and Ayu Bala (lifespan quality/quantity).

#	Parameter	What It Assesses	Assessment Approach
1	Prakriti	Innate constitution (Vata/Pitta/Kapha predominance)	Baseline from which abnormalities are measured; indicates inherent strength.
2	Vikriti	Current pathological deviation from Prakriti	Which doshas are aggravated/lodged and where; the vitiation to treat.
3	Sara	Tissue quality/essence and vitality	Strength of each dhatu (Rasa, Rakta, Mamsa, Meda, Asthi, Shukra, Oja); indicates
4	Samhanana	Body compactness and integration	Tightness/firmness of tissues and joints; lower compactness = more disease pron
5	Pramana	Body measurements and proportions	Anguli (dimensions), Anjali (tissue quantity), Sankhya (organ count).
6	Satmya	Adaptability / lifestyle compatibility	Foods, activities, and behaviors the body has adapted to over time.
7	Sattva	Mental strength and fortitude	Graded Pravara (strong), Madhyama (moderate), Avara (weak).
8	Ahara Shakti	Digestive / metabolic capacity	Appetite, intake patterns, digestive efficiency, metabolic outcomes.
9	Vyayama Shakti	Exercise capacity and endurance	Physical activity tolerance without fatigue/breathlessness.
10	Vaya	Age and life-stage physiology	Contextualizes all other parameters for treatment intensity.

Clinical significance: Kapha-predominant Prakritis are strong, Pitta moderate, Vata less strong. Administering intensively potent medicines without judging tolerance can cause serious, life-threatening complications; giving weak medicines to strong patients with serious disease fails to resolve it. The tenfold examination calibrates treatment intensity to each individual patient.

3. Standard Clinical Note Formats (Output Schema Reference)

The SOAP note (Subjective, Objective, Assessment, Plan) is the dominant clinical documentation format, from Lawrence Weed's problem-oriented medical record (POMR). For MediKiosk Module C, SOAP provides the physician-ready output structure.

Section	Contents	MediKiosk Mapping
S - Subjective	CC + HPI + past medical/surgical + medications + allergies	Family/social history captured by Module A (voice + touch)
O - Objective	Vitals, physical exam, labs, imaging, measurements	Digitized lab values and investigation results from Module B
A - Assessment	Diagnosis + differentials + likely etiologies	AI-generated draft - physician edits/confirm, never auto
P - Plan	Diagnostic workup, treatment, referrals, education, disposition	Set by physician at consultation; kiosk pre-populates draft

The HPI supports several elicitation mnemonics - OLD CARTS (Onset, Location, Duration, Character, Aggravating/Alleviating, Radiation, Temporal pattern, Severity), OPQRST, LOCQSMAT, and SOCRATES. For Medikiosk, SOCRATES (elicitation) and OLD CARTS (documentation) capture overlapping data reorganized for different purposes - the schema should store the underlying structured fields once and render them differently per purpose.

4. Architecture Implications for MediKiosk

- Two-layer schema is required

SOCRATES is symptom-focused (8 fields); Dashavidha Pariksha is person-focused (10 parameters). Both layers must co-exist in the clinical ontology - one does not replace the other.

- Dialogue manager state machine

The full interview sequence (open -> SOCRATES -> red flags -> PMH -> drugs -> allergies -> family/social -> ICE -> summary) defines the dialogue manager's state transitions for Module A.

- Sattva guides UX accessibility

Dashavidha's Sattva grading (Pravara/Madhyama/Avara) maps directly to UI decisions - low Sattva patients need more audio prompts and simpler touch interfaces, aligning with the low-literacy accessibility requirement.

- Output schema from SOAP

Module C's physician-ready summary should follow SOAP: Subjective = structured history, Objective = digitized findings, Assessment = AI draft, Plan = physician-authored.

- Red-flag detection rules

Thunderclap headache, acute chest pain with dyspnoea, and stroke symptoms should trigger priority triage - mapped from the SOCRATES fields (sudden onset + severity + associations).

5. Research Pulse - What the Community Is Saying

- Competing SIH team is active

rohit-h11/medikiosk-sih-26047 has drafted clinical-ontology schemas (issue #2) for both SOCRATES and Dashavidha Pariksha / CCRAS Prakriti Assessment Scale (PAS), and a SOCRATES adaptive LLM dialogue engine (issue #7). Worth studying before building from scratch.

- AI clinical documentation is a live market

Kith (Show HN, Aug 2026) generates AI clinical notes for therapists from ambient session audio - passive capture contrasts with MediKiosk's active questioning approach, validating the broader market need.

- Clinical research profession is stressed

r/clinicalresearch threads on burnout ('CR really is a complete shit show', 224 upvotes) and the NIH ending budding-clinical-researcher grants (160 pts HN) reflect systemic pressures that digital procurement in OPDs partly aims to relieve.

- Reddit was partially coverage-limited

Reddit returned 13 threads before rate-limiting (HTTP 429); r/medicine, r/Ayurveda, r/clinicalresearch were discovered but full-depth extraction was incomplete in this window.

Sources: *Geeky Medics, ClinicalBridge, EasyAyurveda, Wikipedia (SOAP note), ayurvedhealing.com, ijapr.in, ayushdhara.in, ResearchGate, github.com/rohit-h11/medikiosk-sih-26047, Hacker News, r/clinicalresearch, r/Ayurveda, r/medicine. Full evidence saved to last30days raw file.*